Mukul Rayana

AI Engineer | Forward Deployed Engineer

667-646-0719 | ai.mukulray@gmail.com | LinkedIn | GitHub | Portfolio | Open to relocation

Summary

AI engineer who builds agent and retrieval systems that real people use. Won 1st place at the UMD Smith Agentic AI Challenge, runs a public RAG service with live users, and cut a production model's latency 3x as the only ML engineer on an XR team. Good at turning a vague client ask into a shipped fix.

Publication

Voice-Driven Panoramic Imagery: Real-Time Generative AI for Immersive Experiences | IEEE Xplore IDCIoT 2024

- Co-author. Built speech-to-360-degree image generation with Whisper and Stable Diffusion at **2.3x** throughput.

Experience

BeeBox Studios Pvt Ltd | IIT Madras Research Park Apr 2023 - Apr 2024
AI Research Intern Chennai, India

- Turned a loose "make it faster" request into a 3x latency cut for field technicians on XR headsets, from **185 ms to 62 ms** P95, with ONNX export and INT8 quantization of a BERT question-answering model (**83.2% F1**).
- Delivered client-requested multi-peer WebRTC a day ahead of a four-day deadline after taking over an unfamiliar JavaScript and Python module, then tested it end to end on a live call.
- Connected Python inference to the Unity/C# client over TCP as the only ML engineer, and deployed to AWS EC2 with Docker and GitHub Actions CI/CD, supporting **six production releases**.
- Proposed a local model over a paid LLM API to cut token cost and keep documents in-house, which the team later adopted.

Education

University of Maryland, College Park | *M.S., Applied Machine Learning* Aug 2024 - May 2026
SRM Easwari Engineering College | *B.Tech., Artificial Intelligence & Data Science* Dec 2020 - Apr 2024
Certifications: Data Structures & Algorithms Specialization, UC San Diego; IBM Data Science Professional Certificate.

Projects

PharmaChain: Cold-Chain Shipment Incident Response | Live Demo | GitHub Apr 2026

- Won **1st place ($4,000)** in a team of five. Eight agents take a shipment alert to an approved action plan in **12 to 15 seconds**, versus the usual 6 to 8 hours.
- Built the Route and Triage agents, their REST API and an LLM router that fails over across four providers to a rule-based mode, and designed the insurance step to always require human approval.

Irminsul: Public Q&A Service with Cited Answers | Live Demo | GitHub Nov 2025 - Present

- Built and run a service used by **15 Genshin Impact players**: **210 requests/s at 0.45 s p95** with zero errors in a 50-user load test, right source in the top 8 for **94.6%** of test questions, and **113 of 118** live tests passing, including all 19 safety probes.

Promise Ledger: Tracking Housing Promises in City Deals | Live Demo Nov 2026

- Led design and built with AI coding tools, in a team of two for GDG AI for Good 2026, an app that pulls housing promises from city deal PDFs, ties each to its exact quote and page, and leaves the final call to people.
- Code keeps a quote only on a word-for-word page match: **96.3% precision** and **98.1% recall** on a 53-claim development set, about **$0.002 per PDF** on Gemini, **258 tests**, on Google Cloud Run.

RECON: Stale Evidence Detection for Research Search | Live Demo | GitHub Oct 2025 - Jan 2026

- Built a four-agent LangGraph pipeline that catches **44%** of stale claims on a 130-question benchmark, versus **0%** for single-pass RAG, with 2% false positives.

Technical Skills

Languages & Tools: Python, C#, SQL; data structures and algorithms; AI coding assistants
AI/ML: Generative AI, RAG, agentic systems, LangGraph, NLP, QLoRA, PyTorch, TensorFlow, scikit-learn
Integration & Deployment: REST APIs, FastAPI, Docker, GitHub Actions CI/CD, AWS (EC2, S3), Google Cloud Run
Data & Search: PostgreSQL, MongoDB, SQLite, pgvector, Pinecone, FAISS